

Model Training - Conversion

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After completing the dataset annotation tutorial content, we can use the mainboard to start training the model.

This tutorial only covers CLI model training and conversion. You can refer to the official website to modify Python examples.
Due to the performance limitations of the RDK S600 mainboard, model training takes a long time, please be patient.

1、 Model Training

Use CLI command to train the model directly: Copy the yolo26n.pt file to the configuration file directory, then open the terminal in the configuration file directory:

```
cd /home/sunrise/ultraanalytics26/data/yahboom_data/orange_data
```

```
yolo detect train data=orange.yaml model=yolo26n.pt epochs=100 imgsz=640
```

`data`: Dataset configuration file

`model`: Pre-trained model file

`epochs`: Number of training epochs

`imgsz`: Input specified image size

```
File Edit View Terminal Tabs Help
ls... 20 images, 0 backgrounds, 0 corrupt: 100% 20/20 138.9it/s 0.1
s
val: New cache created: /home/sunrise/ultralytics26/data/yahboom_data/orange_data/val/labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and
determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 114 weight(decay=
0.0), 126 weight(decay=0.0005), 126 bias(decay=0.0)
/usr/local/lib/python3.10/dist-packages/matplotlib/projections/__init__.py:63: U
serWarning: Unable to import Axes3D. This may be due to multiple versions of Mat
plotlib being installed (e.g. as a system package and as a pip package). As a re
sult, the 3D projection is not available.
  warnings.warn("Unable to import Axes3D. This may be due to multiple versions o
f ")
Plotting labels to /home/sunrise/ultralytics26/data/yahboom_data/orange_data/run
s/detect/train3/labels.jpg...
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to /home/sunrise/ultralytics26/data/yahboom_data/orange_data/run
s/detect/train3
Starting training for 100 epochs...

Epoch    GPU_mem    box_loss    cls_loss    dfl_loss    Instances    Size
```

2、 Model Conversion

The final model will be generated in the runs folder: Generally select the best.pt file for use

```
/home/sunrise/ultralytics26/data/yahboom_data/orange_data/runs/detect/train/weights
```

Convert pt model to ncnn model:

```
cd /home/sunrise/ultralytics26/data/yahboom_data/orange_data/runs/detect/train/weights
```

```
yolo export model=best.pt format=ncnn
```

References

[Model Training with Ultralytics YOLO - Ultralytics YOLO Documentation](#)